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Reframing ME/CFS: toward a unified mechanistic model of chronic post-infectious diseases

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Abstract

Background Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a severe multisystem illness marked by post-exertional malaise (PEM), cognitive dysfunction, autonomic disturbance, and impaired physiological resilience. Historically, the absence of validated biomarkers, heterogeneous definitions, and limited investigative capacity have complicated mechanistic interpretation and contributed to the use of psychosocial and rehabilitative frameworks in clinical practice and in parts of the literature.

Main body Advances in systems biology, accelerated by Long-COVID research, have transformed our understanding of post-infectious syndromes, implicating persistent immune dysregulation, mitochondrial and metabolic reprogramming, endothelial and microvascular dysfunction, abnormal coagulation, lipid-mediated signalling, extracellular vesicle communication, and viral protein-associated immune activation. This review charts the shift from early post-infectious observations through psychosocial dominance to contemporary biological frameworks, emphasising that pathology is state-dependent and revealed under physiological stress.

Conclusion ME/CFS is thus reframed here as a disorder of impaired adaptive capacity within post-infectious disease biology.

Keywords Post-Exertional Malaise (PEM), Post-infectious disease biology, Immune dysregulation, Mitochondrial and metabolic reprogramming, Impaired physiological resilience

Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a severe multisystem illness characterized by post-exertional malaise, cognitive dysfunction, autonomic disturbance, and impaired physiological resilience [1]. ME/CFS imposes a substantial burden on individuals and society [2–6]. Recent estimates place prevalence at approximately 0.42% of adults in the USA [5], with high-income-country estimates generally ranging from 0.2 to 1%, although true prevalence is likely underestimated because of underdiagnosis and case-definition variability. A recent study [6] reported a prevalence rate for ME/CFS in Germany of 0.78% and 1.7% for the combined number of post-infectious conditions, equating to an annual combined €64.4 billion cost in 2025, equivalent to approximately 1.44% of national GDP.

The condition markedly impairs daily activity, employment, and social participation; a recent scoping review [5] found annual per-patient costs ranging from $2,916

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to $119,611 (U.S), with indirect costs and productivity losses forming the largest component. Health-related quality of life is consistently and profoundly reduced across physical functioning, role limitation, vitality, pain, social participation and general health domains, with impairments comparable to those reported in the post-acute COVID-19 condition (PASC). Patient-reported healthcare experience remains poor. In a large UK survey [3], ME/CFS diagnosis was frequently delayed, with 12.9% waiting more than 10 years, only 10.1% reporting a positive impact of updated NICE guidance on care, and NHS satisfaction remaining low. These data emphasize that ME/CFS is not only a mechanistic research challenge, but also a high-burden clinical condition requiring earlier diagnosis, coordinated multidisciplinary care, and biologically informed therapeutic development.

Despite affecting millions worldwide, ME/CFS has long occupied an uncertain position in medicine. This has been driven, in part, by the frequent absence of consistent abnormalities on routine resting investigations and by the use of heterogeneous case definitions, both of which have complicated patient stratification and reproducibility across studies. In this context, psychosocial explanatory models were widely adopted within parts of the literature and clinical practice. However, these models have been increasingly questioned as methodological limitations have been highlighted and as a growing body of biomedical evidence has demonstrated substantial physiological impairment, a high burden of disease and unmet healthcare need.

This review examines how, in the context of limited mechanistic resolution, pragmatic diagnostic and management approaches have coexisted with ongoing efforts to establish a biologically grounded understanding of ME/CFS. Advances in systems biology, immunology, vascular science, and metabolic profiling, accelerated by Long-COVID research, now reveal a coherent pattern of post-infectious immune-metabolic dysregulation marked by impaired adaptability and stress-revealed dysfunction. This review traces the evolution of ME/CFS from early post-infectious observations, through a period in which mechanistic uncertainty shaped interpretive frameworks, to contemporary biologically informed models, emphasising state-dependent pathology and its implications for experimental design, clinical interpretation, and the ongoing refinement of clinical understanding.

Literature identification and selection

This review is based on a structured narrative synthesis of the literature. Relevant studies were identified through searches of the PubMed database between 1st January 2000 and 30th April 2026, with priority given to primary research studies from the past 6 years.

Search terms included combinations of: "MECFS", "Chronic Fatigue Syndrome", "Long-COVID", "PASC", "immune dysregulation", "mitochondrial dysfunction", "endothelial dysfunction", "metabolomics", "extracellular vesicles" and "Therapeutic".

Priority was given to human cohort studies, mechanistic investigations, multi-omic analyses, and studies incorporating physiological challenge paradigms. Where appropriate, earlier foundational studies were included for context. Given the heterogeneity of case definitions and methodologies, findings are interpreted with attention to variability, cohort dependency, and reproducibility.

Early clinical observations and post-infectious framing

Descriptions of illness consistent with ME date back to the mid-twentieth century, notably in epidemic outbreaks such as the Royal Free Hospital outbreak in London (1955) [7, 8]. These early accounts documented prolonged post-infectious disability with neurological and autonomic features, establishing an initial biomedical and infectious framing. However, the absence of definitive laboratory abnormalities and the episodic nature of outbreaks allowed alternative interpretations to gain traction. Reclassification [9] of the "Royal Free" and other outbreaks, as mass psychogenic illness, introduced an enduring scepticism that would later shape funding priorities and research design.

Case definitions and the expansion of heterogeneity

The introduction of the term "Chronic Fatigue Syndrome" (CFS) in the 1980s and the subsequent CDC case definitions from 1988 and 1994 [10, 11] represented attempts to standardise research cohorts. While these definitions improved reproducibility across studies, they also broadened inclusion criteria and did not require post-exertional malaise (PEM) as a core feature. This had profound downstream effects. Biologically distinct subgroups were combined, diluting signal detection and complicating the interpretation of negative findings. Crucially, subsequent criteria [12, 13], recentred PEM and multisystem dysfunction as hallmarks of the disease.

Evolution of clinical frameworks in the context of limited mechanistic resolution

The biopsychosocial model of illness, originally proposed by Engel, conceptualises health and disease as arising from interactions between biological, psychological, and social factors, and has been widely adopted as a general framework in areas such as rehabilitation and chronic disease management [14]. In contrast to the traditional biomedical model, which seeks discrete pathological causes within the body, the biopsychosocial approach emphasises multi-level influences on illness experience and outcome, including personal context and social environment. Importantly, it was conceived as an extension of biomedical reasoning rather than a replacement, with biological processes remaining a central component of the model. However, the model is inherently non-specific and has been subject to criticism regarding its operationalisation and potential for misapplication, particularly where it is used in the absence of clearly defined biological mechanisms.

In the absence of validated biomarkers, psychosocial models [15] of "Chronic Fatigue" gained prominence, particularly in the UK. These models conceptualised symptoms as perpetuated by maladaptive beliefs, deconditioning, and behavioural responses, and proposed cognitive behavioural therapy (CBT) and graded exercise therapy (GET) as rehabilitative treatments. Within the U.K. primary care system, where the operational demand was for guideline-backed interventions and a service pathway, the prescription of these interventions gained legitimacy when NICE published guidance in 2007 [16]. The publication of the PACE trial in 2011 [17, 18] further legitimised their use, despite later methodological critiques [19, 20] of outcome switching, reliance on subjective outcomes in unblinded settings, and claims of recovery.

The combination of professional consensus, guideline endorsement [16], and the practical ease of service implementation contributed to the sustained use of these approaches within clinical and research settings, even as mechanistic understanding remained limited. Once a psychosocial explanatory model and its associated interventions were embedded within clinical pathways, they became resistant to challenge. In such circumstances, methodological critique was often reframed as controversy rather than treated as a routine component of scientific self-correction. For patients, the consequences were substantial. The elevation of interpretive authority over evidential scrutiny limited access to alternative explanatory models, constrained clinical curiosity, and, in some cases, exposed individuals to interventions that failed to accommodate the biological reality of post-exertional symptom exacerbation.

Methodological reappraisal and the role of external scrutiny

From the mid-2010s onward, sustained methodological critique [19, 20], driven by patients, independent researchers, and investigative journalism, forced a re-examination of the evidential basis of CBT/GET. Despite strong opposition from the PACE trial authors [21], Freedom of Information tribunal rulings, reanalyses of trial data, and peer-reviewed reassessments progressively undermined their claims of robust efficacy. This process culminated in the 2021 revision of UK clinical guidance, which removed GET as a recommended treatment and reframed CBT as supportive rather than curative [22–25]. ME/CFS thus became a rare example of post-hoc guideline reversal driven by evidential reassessment rather than therapeutic innovation.

This history reinforces a broader lesson for medicine. In conditions characterised by uncertainty and unmet need, the availability of an explanation should not be conflated with its validity [20, 26]. Safeguarding patients requires that authoritative claims remain continuously accountable to evolving evidence [27], and that the absence of immediate biomedical answers is not resolved by prematurely settling on explanatory models that are convenient, scalable, or professionally reassuring.

Chronic underfunding, long-COVID, and mechanistic convergence

The longstanding absence of definitive biomarkers or targeted therapies in ME/CFS likely reflects a persistent mismatch between disease burden and biomedical investment [5]. For decades, research into ME/CFS lacked the funding, and therefore the capacity to support large, well-phenotyped cohorts or to test systems-level hypotheses. The emergence of Long-COVID [28] provided a decisive counterfactual. Within months of the onset of the SARS-CoV-2 pandemic, unprecedented funding, infrastructure, and interdisciplinary efforts were mobilised to study post-acute sequelae. This investment did not merely generate SARS-CoV-2-specific insights; it also enabled more detailed investigation of biological processes thought to be relevant to ME/CFS that had previously been limited by smaller study sizes and methodological constraints. (Fig. 1). Consequently, across multiple large-scale Long-COVID cohorts, investigators have independently identified: (a) persistent innate [29–31] and adaptive immune dysregulation [32, 33], including altered monocyte and T-cell states [34, 35]; (b) endothelial dysfunction [36–38] and impaired microvascular regulation [39, 40]; (c) aberrant coagulation and fibrinolysis-resistant fibrin(ogen) aggregates [41, 42]; (d) metabolic reprogramming and mitochondrial stress signatures [43]; (e) circulating pathogenic factors capable of inducing cellular dysfunction in vitro [44, 45]; (f) evidence of viral persistence [46] or abortive reactivation without productive infection [47, 48]. Each of these domains maps directly onto pre-existing strands of ME/CFS research. What Long-COVID has provided is not conceptual novelty but statistical power, temporal anchoring, and institutional legitimacy.

Long-COVID studies also showed that chronic post-infectious illness can follow mild acute infection [49], challenging assumptions that persistent symptoms may reflect premorbid vulnerability or psychological traits. This parallels longstanding ME/CFS observations after Epstein-Barr virus, enteroviruses, and other infections, reinforcing a unified post-infectious framework. Sociologically, Long-COVID has functioned as a natural experiment in scientific attention. When modern tools, such as single-cell transcriptomics [50], high-resolution metabolomics, endothelial imaging and extracellular vesicle profiling [51], were applied at scale, biological signals rapidly emerged [52]. The implication for ME/CFS is direct: prior mechanistic uncertainty reflected insufficient investigative capacity, not the absence of biology. Long-COVID has therefore advanced ME/CFS research by validating certain core biological concepts under adequate investment. Convergent findings across immune, vascular, metabolic, and virological domains [45, 53], indicate that ME/CFS belongs within a broader class of post-infectious, immune-mediated chronic diseases rather than representing an exceptional entity.

Mechanistically, recent Long-COVID research [42] suggests that abnormal fibrin architecture and impaired fibrinolysis can be driven by circulating soluble factors such as cytokines, metabolites, and acute-phase proteins, even when fibrin derives from healthy-donor fibrinogen, supporting a model in which post-infectious plasma transfers prothrombotic behaviour independently of conventional coagulation measures.

Recently, Mignolet et al. [54] provide evidence that purified IgG derived from long-COVID patients with neurological sequelae can induce pain-related phenotypes following passive transfer into mice, with preferential localisation to dorsal root ganglion sensory neurons and loss of effect following IgG depletion or enzymatic cleavage. These results have been largely corroborated by many independent studies [55, 56]. While these findings do not imply mechanistic equivalence between Long-COVID and ME/CFS, nor account for the full spectrum of clinical manifestations, they substantiate the more circumscribed proposition that post-infectious chronic illness may, in part, be mediated by circulating immune effectors with demonstrable functional activity. This convergence also has implications for research governance: the pace of progress on Long-COVID demonstrates what is achievable when disease burden, funding, and methodological ambition align. For ME/CFS, these developments are consistent with long-reported patient experiences and provide an emerging basis for the translation of mechanistic insights into more targeted therapeutic approaches.

Scientific advances and the return of emphasis on biology

(A) Physiological challenge, systems-level measurement, and the emergence of coherent biological signals

The refinement of two-day cardiopulmonary exercise testing (CPET) [57, 58] protocols represented a pivotal methodological advance in ME/CFS research. By demonstrating reproducible impairments in recovery following exertion, manifesting as reduced oxygen consumption, workload, and anaerobic threshold on repeat testing, these studies [59, 60] provided objective evidence that PEM may reflect a state-dependent physiological failure rather than deconditioning or effort-related variability. This work established exertional challenge as a critical lens through which core disease biology becomes visible.

The development of large-scale deep phenotyping initiatives [61, 62] enabled the integration of immunological [63], neurological, metabolic, and vascular assessments within the same patient cohorts. These multimodal datasets moved the field beyond isolated, single-candidate hypotheses and instead revealed coordinated abnormalities spanning multiple physiological systems. Importantly, such approaches demonstrated that no single pathway operates in isolation, reinforcing the need for integrative interpretation.

Advances in systems biology and metabolic profiling further consolidated this shift [64–68], revealing a hypometabolic signature, characterised by alterations in redox balance, mitochondrial function, and lipid-mediated signalling. When considered alongside exercise physiology and deep phenotyping data, these findings support a model in which impaired metabolic adaptability and system-wide coordination underlie the characteristic intolerance to physiological stress observed in the disease.

(B) Mitochondria as orchestrators of metabolic and immune reprogramming

For much of modern medicine, mitochondria were viewed chiefly as passive bioenergetic organelles, the cell's "powerhouses", relevant mainly to ATP production and rare inherited disorders. This reductionist perspective proved limiting in complex, multisystem illnesses such as ME/CFS, encouraging the assumption that, without overt mitochondrial failure, cellular energetics were intact and persistent symptoms must arise from central or behavioural causes. Over the past decade, this view has been fundamentally revised [69]. Mitochondria are now recognised as master regulators of cellular homeostasis, integrating metabolic flux, redox balance, innate immune signalling, calcium handling, apoptosis, epigenetic control, and intercellular communication. Rather than isolated generators, they function as dynamic signalling hubs sensing stress and coordinating system-wide responses, an essential shift for interpreting ME/CFS biology and understanding why earlier frameworks failed to capture its complexity.

A key inflection point was Robert Naviaux's "Cell Danger Response" (CDR) [70], an evolutionarily conserved metabolic and signalling state triggered by infection, injury, or toxic stress. In this model [71], mitochondria actively reprogramme metabolism away from growth towards defence, repair, and containment,

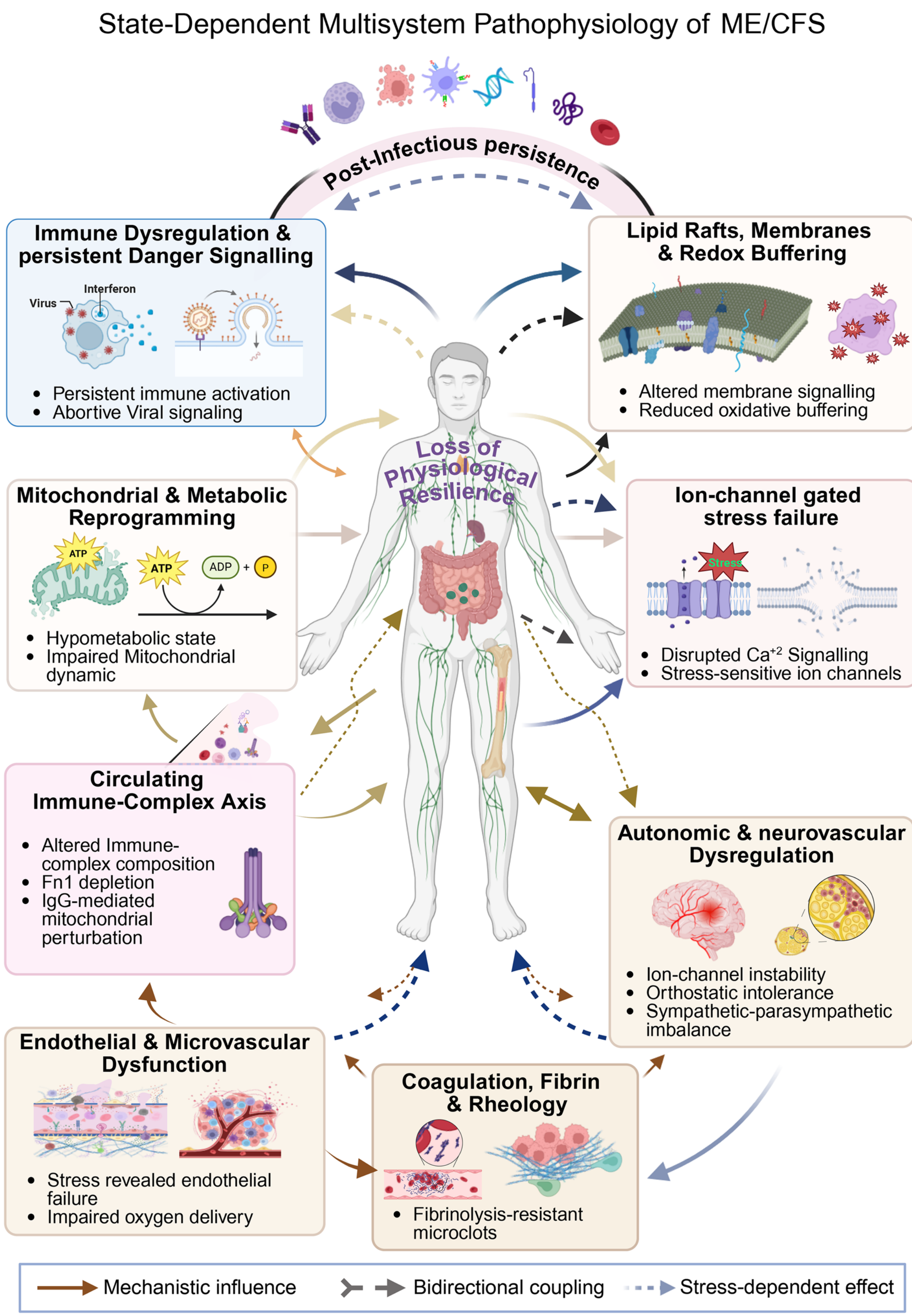

Fig. 1 (See legend on next page.)

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Fig. 1 Schematic representation of state-dependent multi-system pathophysiology of ME/CFS. Created in https://BioRender.com. This schematic presents a systems-level model of myalgic Encephalomyelitis/chronic fatigue syndrome (ME/CFS) in which multiple biological domains converge upon a shared phenotype of loss of physiological resilience. Solid arrows denote mechanistic influence; dashed bidirectional arrows indicate reciprocal coupling; dotted arrows represent state-dependent effects that become clinically prominent under exertional or orthostatic stress rather than at rest. Persistent post-infectious molecular signalling potentially provides the overarching context. Immune dysregulation and persistent danger signalling, including abortive viral reactivation, likely occupy an upstream position and appear to be functionally linked with lipid raft organisation and redox buffering, consistent with the interdependence of immune receptor signalling, membrane microdomains, and oxidative tone. Mitochondrial and metabolic reprogramming, characterised by hypometabolic signatures and impaired mitochondrial dynamics, may constrain adaptive capacity. Bidirectional immune-metabolic coupling provides a framework through which inflammatory signalling reshapes metabolism, while mitochondrial redox state may, in turn, modulate immune activation. A distinct circulating immune-complex axis (altered immune-complex composition, fibronectin depletion, IgG-mediated mitochondrial perturbation) potentially functions as a mechanistic bridge, linking immune disturbance to endothelial and mitochondrial vulnerability. Endothelial and microvascular dysfunction is depicted as stress-revealed, reciprocally coupled with coagulation, fibrin architecture, and rheology, highlighting thrombo-inflammatory feedback within the microcirculation. Ion-channel-gated stress failure and autonomic/neurovascular dysregulation likely represent downstream expressions of systemic vulnerability, with reciprocal interactions governing vascular tone and orthostatic adaptation

altering oxidative phosphorylation, purine signalling, lipid metabolism, and redox balance, adaptive short term but pathological if chronically sustained [72]. Naviaux's metabolomic studies in ME/CFS [73] provided the first systems-level evidence of a coordinated hypometabolic state across multiple pathways rather than a single enzymatic defect, reframing "low energy" as an actively maintained cellular condition consistent with persistent danger signalling and suggesting ME/CFS may represent a stalled recovery programme.

Parallel work [74–76] has extended mitochondrial biology into virology, immunology, and vascular function, demonstrating that factors in patient sera, particularly immunoglobulin-associated components, can induce mitochondrial fragmentation, impaired respiration, and redox stress in healthy cells. This links circulating immune phenomena to mitochondrial dysfunction and supports the view of mitochondria as both targets and amplifiers of immune dysregulation, occurring in the absence of overt viral replication or intrinsic mitochondrial genetic defects. Furthermore, mitochondrial dynamics, membrane potential, and metabolite signalling shape innate immune responses [77, 78], endothelial behaviour [79], and inflammatory signalling cascades [80], providing a unifying mechanism by which post-infectious immune disturbance may produce systemic symptoms, impaired oxygen utilisation, and exertion intolerance [81].

A systematic reanalysis of transcriptomic and proteomic datasets by Keele et al. [82] highlights a fundamental limitation in the current evidence base: reproducibility at the level of individual genes is minimal, with only two genes consistently replicated across studies, and proteomic convergence largely absent. These findings underscore the heterogeneity of ME/CFS and the limitations imposed by small sample sizes, variable study designs, and tissue-specific effects. Importantly, however, the study identifies convergence at a higher level of organisation, with mitochondrial dysregulation emerging as a recurrent feature across multiple data modalities. This suggests convergence at the level of biological systems, despite limited reproducibility at the level of individual molecular signals.

Also notable is accelerated biological ageing reported in ME/CFS [83], including premature telomere attrition in peripheral blood cells, consistent with chronic cumulative cellular stress. A recent mechanistic review [84] proposes that long-term interaction with tissue-resident pathogens can drive core ageing pathways through persistent immune activation, mitochondrial disruption, redox imbalance, and senescence-linked signalling, linking infection biology to "inflammageing" phenotypes relevant to chronic post-infectious syndromes.

(C) Viral proteins, abortive lytic replication, and immune dysregulation

A pivotal conceptual advance has been the proposal that herpesvirus proteins produced during abortive lytic replication, particularly viral dUTPases, could drive chronic immune activation independent of productive infection [85, 86]. Experimental work has shown that herpesvirus dUTPases [87] can activate innate immune pathways [88, 89], induce pro-inflammatory cytokines, and engage stress signalling cascades. This model resolves longstanding inconsistencies between strong epidemiological links to herpesviruses and repeated failure to demonstrate active replication.

However, while herpesviruses have provided the most experimentally tractable model for this phenomenon, the conceptual implications extend beyond a single viral family. A growing body of evidence suggests that abortive or incomplete viral reactivation, characterised by restricted expression of immunostimulatory viral proteins, and without production of infectious virions, may represent a common mechanism across multiple persistent or latent viruses [90]. In such states, viral genomes are transcriptionally active [91] enough to generate proteins capable of engaging innate immune sensors [92], yet insufficiently active to trigger classical cytopathic effects or detectable viraemia.

By decoupling immune activation from full lytic replication, abortive reactivation provides a biologically plausible route to chronic immune stimulation, metabolic stress signalling, and endothelial vulnerability in the absence of overt viral load. Importantly, this represents a departure from traditional infection models rooted in Koch's postulates, which assume that pathogenicity requires replicating organisms and tissue invasion. In post-infectious syndromes such as ME/CFS, an element of the pathophysiology may involve persistent or intermittently renewed immune engagement with viral-derived products, generated episodically or at low levels, sufficient to maintain danger signalling but insufficient to provoke sterilising immunity or straightforward diagnostic detection.

(D) Fibronectin-immunoglobulin complexes as a convergent immune-vascular mechanism in ME/CFS and PASC

Recent work [91] provides a mechanistically coherent bridge between systemic immune dysregulation and endothelial/mitochondrial vulnerability in ME/CFS and post-acute sequelae of infection. The key observation is directionally discordant fibronectin (Fn1) biology, with increased circulating Fn1 in serum yet depletion of Fn1 in IgG-bound circulating immune complexes (CICs) from ME/CFS patients. Increased circulating Fibronectin was also observed in the plasma of ME/CFS patients in an independent cohort [39]. This pattern argues against a simple "more Fibronectin everywhere" phenomenon and instead supports a selective alteration in immune-complex composition/handling, i.e., what is available in serum is not being incorporated (or retained) within the antibody-bound complexes that ordinarily participate in immune clearance functions. This matters because fibronectin is not merely a structural extracellular-matrix protein [92]. It also participates in innate immune signalling and opsonisation-like functions through binding to complement components [93] (including C1q and C3), and it can modulate inflammatory responses. We explicitly note that reduced Fibronectin within immune complexes could plausibly imply impaired pathogen clearance and/or impaired scavenging of cellular debris, which in turn would be expected to increase downstream immune activation pressure and diversify autoreactivity over time.

In parallel, elevated circulating Fibronectin is potentially pro-inflammatory [94–97], because circulating fibronectin can bind TLR4 and induce cytokine responses. Notably, specific Fn1 domains are also able to activate platelets and mast cells, three pathways directly relevant to vascular tone/microthrombosis, MCAS-like signalling, and endothelial stress susceptibility in post-infectious syndromes. A further "tightening" of the mechanistic narrative comes from immunoglobulin profiling, where antibodies against fibronectin, considered as natural IgM species [98], with homeostatic scavenger/protector/regulator roles, were observed to be depleted after COVID infection [91]. This is significant because it strengthens the argument that immune perturbation is not solely about the presence of pathogenic antibodies but also about the loss of protective, homeostatic antibody functions that normally constrain inflammation, facilitate immune-complex processing, and limit the escalation of autoimmunity.

These findings integrate closely with our recent functional work, which shows that patient-derived IgG can enter endothelial cells (HUVECs) and, in a subset of patients (notably with a sex-stratified signal), drives mitochondrial network fragmentation and measurable changes in mitochondrial proteins and energetics. In the same report [99], we demonstrated that passive transfer of IgG to healthy PBMCs induced secretion of specific inflammatory cytokines (with IL-1β reaching statistical significance across the cohort, while other cytokines showed patient-subset effects), reinforcing the concept that circulating immunoglobulin fractions can act as functional effectors rather than simply passive biomarkers.

Taken together with the fibronectin/fibronectin IgM antibody observations, we speculate that ME/CFS and PASC may involve (a) altered immune-complex composition and homeostatic antibody depletion plus; (b) immunoglobulin-mediated perturbation of endothelial mitochondrial resilience.

(E) Endothelial dysfunction as a state-dependent failure of physiological adaptability

Accumulated evidence of endothelial dysfunction in ME/CFS [100–104] suggests impaired microvascular regulation and reduced oxygen delivery, particularly under exertional stress. The COVID-19 pandemic accelerated interest in endotheliopathy, as SARS-CoV-2 infection has been shown to induce endothelial injury and thrombo-inflammatory states [105, 106]. The convergence of endothelial dysfunction [107, 108], platelet activation, and coagulation abnormalities has therefore focussed attention on the subject of vascular homeostasis in ME/CFS.

A key conceptual refinement from recent experimental work concerns the conditional nature of endothelial dysfunction in ME/CFS. Contrary to earlier assumptions, endothelial cells exposed to ME/CFS plasma do not appear overtly abnormal under resting, non-stressed conditions [109]. Morphology, baseline viability, and some standard functional measures may remain within reference ranges in quiescent states. However, growing evidence indicates that the primary abnormality lies not in basal endothelial structure but in metabolic responsiveness to stress. When subjected to immune, oxidative, mechanical, or metabolic challenges, approximating physiological exertion, infection, emotional stress, or orthostatic load, endothelial cells exposed to ME/CFS sera show impaired adaptive capacity. Instead of mounting a flexible metabolic response, they are constrained in up-regulating energy production, maintaining redox balance, preserving barrier integrity, and coordinating nitric oxide and mitochondrial signalling. The outcome is a shift from apparent normality to dysfunction that is state-dependent rather than constitutive.

This distinction has major implications for experimental design and interpretation. If endothelial pathology emerges chiefly under metabolic demand, studies that do not induce such demand are inherently unable to detect the relevant dysfunction. Mechanistically, this stress-revealed failure aligns with broader concepts of metabolic reprogramming and the cell danger response. Endothelial cells are metabolically active integrators of immune, vascular, and energetic signals, and their capacity to shift between metabolic states is essential for maintaining perfusion, oxygen delivery, and inflammatory homeostasis during stress. A chronic post-infectious state in which circulating immune factors restrict mitochondrial flexibility or redox adaptability would therefore render endothelial cells functionally vulnerable precisely when physiological demand rises.

Recognising this conditional pathology reframes negative findings and research priorities. It potentially clarifies why resting biomarkers and static imaging may underestimate disease burden and highlights the need for provocative or stress-based assays, including metabolic, inflammatory, or mechanical analysis to meaningfully assess endothelial function in ME/CFS. More broadly, it may reflect a central principle of post-infectious disease biology: when pathology is dynamic rather than structural, it must be studied in motion, not at rest.

(F) Autophagy and mitophagy as regulators of immune-mitochondrial resilience

Further evidence [110] supporting the immune-metabolic framing of ME/CFS concerns autophagy and mitophagy. Autophagy is not merely a housekeeping process but a central regulator of immune-metabolic control, influencing inflammatory signalling, antigen presentation, mitochondrial quality control, and resolution of cellular stress. Notably, Gottschalk et al. reported that ATG13, an autophagy protein essential for autophagosome initiation, was markedly elevated in ME/CFS serum across independent cohorts, and that exogenous ATG13 triggered inflammatory signalling in vitro, consistent with broader disruption of the autophagy programme.

Within a pathogenesis model emphasising persistent post-infectious immune activation, these findings support the proposition that innate immune cells may enter a chronically altered state in which stress-response pathways remain engaged, mitochondrial turnover is inefficient, and inflammatory set-points are shifted. Conceptually, impaired mitophagy or dysregulated autophagy signalling provides a biologically plausible pathway [111] linking infection-triggered immune activation to sustained redox stress, reduced bioenergetic capacity, and amplified inflammatory signalling, without requiring major tissue damage or ongoing productive viraemia.

(G) Lipid organisation and immune signalling

Convergent findings in ME/CFS and Long-COVID implicate membrane lipid organisation, particularly lipid raft composition, in immune dysfunction. Lipid rafts function as signalling platforms that regulate receptor clustering and activation thresholds [112], including Toll-like receptor signalling. Recent ME-focused work [113] identifies SMPDL3B (sphingomyelin phosphodiesterase acid-like 3B), a membrane-associated regulator of sphingolipid balance, as a potential biomarker and mechanistic mediator: altered SMPDL3B may change raft fluidity and receptor organisation, thereby modifying immune responses to inflammatory and antiviral stimuli. This is consistent with post-infectious models in which immune cells display both heightened inflammatory tone and impaired or poorly coordinated antiviral responsiveness. Practically, the lipid raft framework links commonly observed "omics-level" lipid abnormalities in ME/CFS to cellular immunophenotypes, since raft composition depends on sphingolipid and cholesterol handling and can be remodelled by systemic metabolic cues [114] and chronic inflammatory signalling.

(H) Haptoglobin as an oxidative-haemolysis buffer with relevance to PEM

Complementary work from Alain Moreau's group [115] highlights haptoglobin (Hp), in both quantity and proteoform structure, as relevant to core symptoms such as PEM and cognitive dysfunction. Hp scavenges free haemoglobin released during haemolysis and thus buffers against haem-mediated oxidative stress and endothelial injury. Moreau's findings support the concept that reduced functional Hp capacity permits exaggerated oxidative and vascular stress during or after exertion, with downstream effects on microvascular regulation and symptom relapse. Considered alongside evidence of endothelial dysfunction and redox disturbance in ME/CFS, they strengthen a pathway in which exercise or orthostatic stress provokes disproportionate oxidative and inflammatory responses because protective scavenging systems are quantitatively or qualitatively insufficient.

Taken together, ATG13/autophagy perturbation, SMPDL3B/lipid raft regulation, and haptoglobin proteoform variation point toward a common theme: lipid biology is not an adjunct topic in ME/CFS but is potentially a mechanistic substrate that links immune signalling, mitochondrial quality control, redox buffering, and endothelial stability. Lipids provide structure to membranes and substrates for inflammatory mediators. They govern receptor microdomains, whilst regulating mitochondrial dynamics [116]. Consequently, lipid dysregulation could plausibly yield a persistent "mis-set" immune state after infection.

(i) Extracellular Vesicles as Mechanistic Messengers

Extracellular vesicles (EVs), including exosomes, have emerged as a critical interface between cellular stress, immune signalling, and systemic effects [117]. EVs carry proteins, lipids, nucleic acids, and mitochondrial fragments reflective of their cells of origin. In ME/CFS, EV cargo profiling [118] has revealed (a) altered cytokine and proteomic signatures, (b) severity-associated immune cell markers, (c) exercise-induced changes aligned with PEM [119], and (d) the presence of mitochondrial DNA capable of activating innate immune pathways. EV biology therefore provides a plausible mechanism for how local cellular dysfunction (immune, metabolic, or endothelial) can propagate systemic effects without overt tissue damage or high-level viraemia.

(j) Selective immune cell reprogramming and organism-level physiological disruption

Recent advances in single-cell and functional immunology have sharpened the understanding of how specific immune cell populations can exert disproportionate effects on whole-organism physiology in post-infectious disease. Rather than reflecting uniform immune activation, ME/CFS and Long-COVID increasingly appear to involve selective, stable reprogramming of particular immune subsets, most notably monocytes [120–122] and T cells [46, 123–125]. Studies have identified distinct monocyte transcriptional and metabolic states in Long-COVID [126], characterised by altered interferon signalling, inflammatory tone, and metabolic gene expression. Such monocytes are well-positioned to influence vascular function, coagulation, and tissue oxygenation through cytokine release, endothelial interaction, and modulation of redox balance. These findings resonate closely with earlier ME/CFS observations of innate immune activation and provide a cellular substrate for sustained endothelial and metabolic stress.

In parallel, recent work by Kumar et al. [35] has demonstrated persistent alterations in monocyte activation profiles following viral infection, with consequences for immune regulation, mitochondrial signalling, and systemic energy balance. Importantly, these changes can persist long after viral clearance and may constrain immune flexibility, thereby amplifying inflammatory responses to secondary stressors such as exertion or infection. Complementing monocyte-state findings, studies report functional impairment of antiviral T-cell responses in ME/CFS [127] and Long-COVID [128], including $CD8^{+}$ T-cell dysfunction with exhaustion-like features and altered cytokine production, alongside expansion of atypical $CD4^{+}$ $CD8^{+}$ 'double-positive' T-cell populations. These observations strengthen the view that selective immune cell reprogramming (rather than uniform inflammation) may contribute to impaired pathogen control and maladaptive stress responses at the whole-organism level. Recent proteomic analyses [129] further suggest that these patterns may reflect a more persistent reconfiguration of immune signalling, characterized by stable shifts in immune-cell-derived outputs rather than transient activation states.

Single-cell studies strengthen, but also qualify, the comparison between Long-COVID and ME/CFS. A recent study [130] reported persistent immune dysregulation in prolonged Long-COVID, with altered lymphocyte subset proportions, cytotoxic and exhaustion-associated gene expression, and impaired intercellular communication persisting up to 1.5–2 years after acute infection. Complementing this, Shahbaz et al. [131] examined Long-COVID patients meeting ME/CFS criteria and identified broad immune remodelling involving monocytes, NK cells, T cells, B cells, platelets, low-density neutrophils, MAIT cells and γδ T cells. Importantly, comparison with idiopathic ME/CFS suggested both overlap and divergence: shared immune activation was evident, but Long-COVID-associated ME/CFS showed more pronounced T-cell exhaustion, NK-cell alteration, MAIT/γδ T-cell depletion and inflammatory monocyte skewing. These findings support the placement of ME/CFS and Long-COVID within a broader post-infectious disease framework, while arguing against unqualified mechanistic equivalence. When integrated with evidence of circulating immune complexes, mitochondrial vulnerability, and stress-revealed endothelial dysfunction, these immune cell findings support a model in which post-infectious disease is maintained by maladaptively reprogrammed immune populations. In this framework, immune cells function not merely as effectors of inflammation, but as mobile regulators of metabolic and vascular physiology, capable of shaping organism-level responses to stress.

(K) Autoimmunity, ion channel dysfunction, and calcium signalling as stress-revealed mechanisms

In parallel with advances in systems biology and vascular research, a substantial body of work [132–136] has drawn attention to immune-mediated and ion-channel level dysfunction in ME/CFS, further reinforcing the view that pathology is revealed under conditions of physiological stress rather than at rest. Notably, studies [133, 137] have identified functional autoantibodies, particularly against β-adrenergic and muscarinic receptors, that are capable of perturbing vascular tone, autonomic regulation, and immune signalling.

At the cellular level, dysfunction of the TRPM3 calcium-permeable ion channel [138], particularly in natural killer cells, has been revealed. Impaired TRPM3 activity [139] provides a direct mechanistic link between altered calcium signalling, immune cell activation, and defective stress responses. Calcium flux is central to mitochondrial function, redox regulation, and endothelial responsiveness; thus, ion-channel abnormalities offer a molecular explanation for how immune and metabolic systems may fail to adapt appropriately when physiological demand increases.

Related observations concerning mast cell activation [140] and exertion-linked calcium accumulation in tissues [141] further support this framework. Mast cells occupy a strategic position at the neuro-immune-vascular interface [142] and are exquisitely sensitive to mechanical, metabolic, and emotional stressors. Dysregulated mast-cell responses can amplify vascular permeability, inflammatory signalling, and sensory disturbance in a manner that may be unapparent under resting conditions but becomes pronounced during stress. Taken together, these lines of investigation do not constitute competing explanations, but rather complementary manifestations of a shared underlying principle in ME/CFS, suggesting that immune, vascular, and metabolic systems retain partial baseline function yet exhibit impaired adaptability when challenged. The convergence of autoantibody research, ion-channel physiology, and calcium signalling with mitochondrial and endothelial biology underscores the need for investigative approaches that probe dynamic responses and further consolidates ME/CFS within a broader class of post-infectious disorders defined by loss of physiological resilience [143].

(L) Moving beyond static, resting-state investigation

Recent research by Bergemann has begun to exploit systemic perturbation paradigms as a means of interrogating immune-metabolic dysfunction in ME/CFS. In a small but mechanistically informative study [144], whole-body hyperthermia was used as a controlled physiological challenge, with immediate pre- and post-intervention analysis of patient-derived peripheral blood mononuclear cells. This approach revealed a consistent reduction in elevated autophagy markers alongside marked increases in mitochondrial respiratory capacity, suggesting that at least some cellular stress signatures in ME/CFS are dynamically modifiable. Importantly, these changes are plausibly attributable not to intrinsic correction of mitochondrial machinery, but to transient improvement in upstream constraints such as tissue perfusion and oxygen availability. As such, the findings support a model in which metabolic reprogramming and sustained stress signalling may arise secondarily from impaired physiological support systems, and they reinforce the broader contention that biologically meaningful abnormalities in ME/CFS are most readily exposed (and interpreted) under conditions of controlled challenge rather than at rest.

Emerging approaches, including time-resolved single-cell transcriptomics (single-cell Thiol-linked alkylation of RNA for sequencing (scSLAM-seq)) [145, 146], now permit direct measurement of nascent, stimulus-responsive transcriptional dynamics in defined immune-cell populations, thereby enabling functional interrogation of cellular adaptability under controlled perturbation rather than static resting conditions. Similarly, cerebrospinal fluid proteomic analysis [147] demonstrated that an exercise challenge unmasks dynamic, post-exertional alterations in neuroimmune and metabolic signalling pathways in ME/CFS that are not evident at rest. Building upon previous work [148], a low-burden challenge strategy is being applied in neuroimaging by Prof. Godlewska (University of Oxford), where functional MR spectroscopy is being used to measure lactate responses during visual stimulation; the underlying premise being those compensatory mechanisms mask abnormalities at rest, whereas activation reveals impaired metabolic adaptability.

Implications of emerging biological insights for clinical interpretation

The increasing recognition of mitochondria as central regulators of cellular signalling carries important implications for the interpretation of complex chronic disease. Earlier explanatory models were developed in a context of limited biological resolution and often relied on the absence of gross structural pathology or single-pathway defects as a basis for inference. Emerging evidence now suggests that subtle, distributed, and potentially reversible alterations in cellular signalling can produce substantial functional impairment without manifesting as discrete lesions detectable by conventional clinical investigation. In this context, ME/CFS may illustrate a broader principle in clinical reasoning: when available investigative frameworks are insufficient to capture underlying biological complexity, there is a risk that poorly characterised processes are interpreted as absence of underlying pathology. At the same time, it is important to recognise that current biological models remain incomplete, with heterogeneity across cohorts and limited reproducibility at the level of specific molecular signals. The growing evidence for interactions between mitochondrial function, metabolic regulation, and immune signalling therefore represents not a definitive resolution, but an evolving framework that may help to reconcile previously disparate observations. As methodological approaches continue to improve, refinement of these models will help to ensure that clinical interpretation remains aligned with emerging biological understanding.

State-dependent dysfunction and the limits of resting clinical inference

A crucial emerging insight [99] is the recognition that key abnormalities in ME/CFS are state-dependent rather than constitutive. Cellular and vascular systems may appear functionally intact under resting conditions, yet fail when exposed to physiological, metabolic, or emotional stress. In this context, normal findings at rest do not signify normal function; they merely indicate that the system has not yet been sufficiently challenged. The failure to account for this distinction has materially shaped clinical interpretation. Conventional medical reasoning, optimised for detecting static lesions or baseline abnormalities, has struggled to accommodate diseases characterised by impaired adaptive capacity. As a result, exertion-triggered symptom exacerbation, which is now recognised as a defining feature of ME/CFS, was often discounted when resting investigations proved unrevealing. From a standpoint of professional learning, this insight underscores the need to revise what is considered diagnostically informative. In conditions defined by loss of physiological resilience, investigative approaches should embrace a recalibration of clinical inference, whereby stress-induced deterioration is regarded as a central biological signal rather than an interpretive difficulty.

Translation to clinical practice

To address the translational implications of emerging ME/CFS biology while avoiding overstatement regarding therapeutic efficacy, we have included a supplementary table (Table S1), summarising candidate interventions for which clinical, mechanistic, or early-phase evidence has been reported. The table distinguishes interventions supported by randomized trial data from those based on preliminary or mechanistic findings, and maps each intervention to the immune, vascular, mitochondrial, autonomic, or redox pathways discussed in this review.

The translational implications of emerging biological insights extend beyond therapeutic development to include diagnosis and clinical management. In the absence of validated biomarkers, diagnosis of ME/CFS continues to rely on clinical criteria, with variability in case definitions contributing to heterogeneity across both research and practice. Recognition of state-dependent pathology has potential relevance for clinical assessment, as symptom expression and physiological abnormalities may vary according to exertional status and disease phase. In current practice, management remains largely supportive and symptom-directed, with emphasis on individualised approaches that take into account functional limitation and post-exertional symptom exacerbation. Multidisciplinary care and improved clinical recognition are therefore important components of patient support while biologically informed diagnostic and therapeutic strategies continue to evolve.

Concluding remarks

The available evidence currently points to ME/CFS being best understood as a disorder of impaired physiological adaptability and resilience. The contemporary literature converges on a biologically coherent model in which immune disturbance, metabolic reprogramming, endothelial dysfunction, and impaired physiological adaptability interact across systems. This reframing positions ME/CFS within a broader class of post-infectious, immune-mediated chronic diseases characterised by dynamic, stress-revealed pathology rather than fixed structural lesions, challenging reliance on resting-state inference and highlighting the need for investigative paradigms that capture conditional dysfunction.

Several core messages follow. Mechanistic convergence across immunological, vascular, metabolic, and virological domains supports a unified systems-level interpretation. The field's history illustrates how evidential asymmetry and operational convenience can shape clinical frameworks when investigative capacity is limited. Advances in Long-COVID research demonstrate that, with adequate scale and interdisciplinary methodology, coherent biological signals become detectable, reinforcing impaired adaptive capacity as a central concept in chronic post-infectious disease biology.

Key uncertainties remain, including the upstream biological events that potentially stabilise persistent immune-metabolic dysregulation, the contribution of viral persistence and immune-complex biology, and the mechanisms linking circulating factors to endothelial and mitochondrial vulnerability. Determining whether these abnormalities are primary drivers or secondary adaptations will require integrated longitudinal and mechanistic investigation. The remaining critical questions are therefore why only a subset of individuals exposed to an acute infectious trigger develop persistent illness, and how such a triggering event (or sequence of insults/events) transitions into a sustained alteration of immune-metabolic-vascular regulation that appears to constrain adaptive capacity and to delay recovery following exertional stress.

The evidence suggests that future progress depends on large, deeply-phenotyped cohorts that incorporate physiological challenge, together with integrated multi-omic and functional approaches. These might usefully include time-resolved single-cell transcriptomics (scSLAM-seq) to characterise dynamic transcriptional responses and support causal inference, alongside prioritisation of plasma-mediated pathophysiology, biomarker development, and mechanistically guided clinical trials. Sustained investment, interdisciplinary collaboration, and methodological transparency will be important in advancing the translation of emerging biological findings into improved diagnostic approaches and the development of targeted therapies. With growing convergence of mechanistic insight and technological capability, there is now a realistic prospect that the field can move toward targeted intervention and evidence-based clinical transformation.

Supplementary Information

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Supplementary Material 1

Author contribution

Conceptualization - PW and BKP contributed equally; Original draft - PW; Review and editing - BKP.

Data availability

Data sharing is not applicable to this article as no new datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable. This study does not require ethical approval and consent of participation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) as a generative AI tool to assist with brainstorming and language refinement. The tool was used to improve clarity, organisation, and readability of the text. No unpublished data were generated or analysed using AI tools, and no AI-generated content was included without critical review. Following its use, the authors carefully reviewed, revised, and edited all content to ensure accuracy, balance, and compliance with journal standards. The authors take full responsibility for the integrity, originality, and accuracy of the final manuscript.

Conflict of interest

The authors declare no conflict of interest.

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